

Time Series Modeling

ARIMA Revenue Forecasting

Research Question

Can the given dataset showing revenue over time be implemented in a prediction model to provide a forecast of revenue for the next quarter?

Analysis Goal

This data analysis aims to use an ARIMA model to output a forecast of revenue for the ninety-day quarter directly after the dataset ends.

Time Series Assumptions

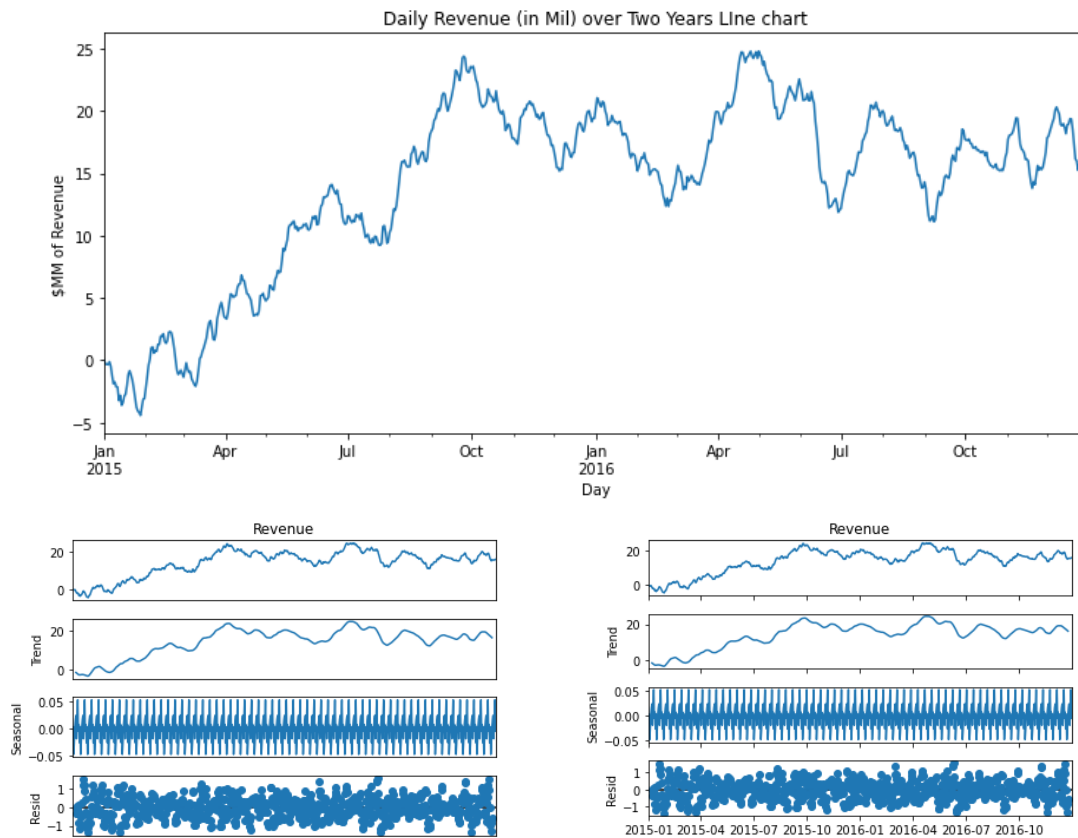
A time series model assumes that data is stationary, meaning long term trends over time have been removed (visually flattened). This model also assumes that autocorrelation remains constant over the relevant time period.

Data Preparation

The data dictionary states that this time series consists of 731 total days, with daily data recordings. Day one records a revenue of zero, but there are no other gaps apparent in the data.

Prior to decomposing the dataset, this data is not stationary. However, when decomposition is implemented, stationarity is achieved, as illustrated by the following visualizations of the transformed dataset. The 'seasonal' graph now shows a consistent range for the data values and residuals show no clear pattern.

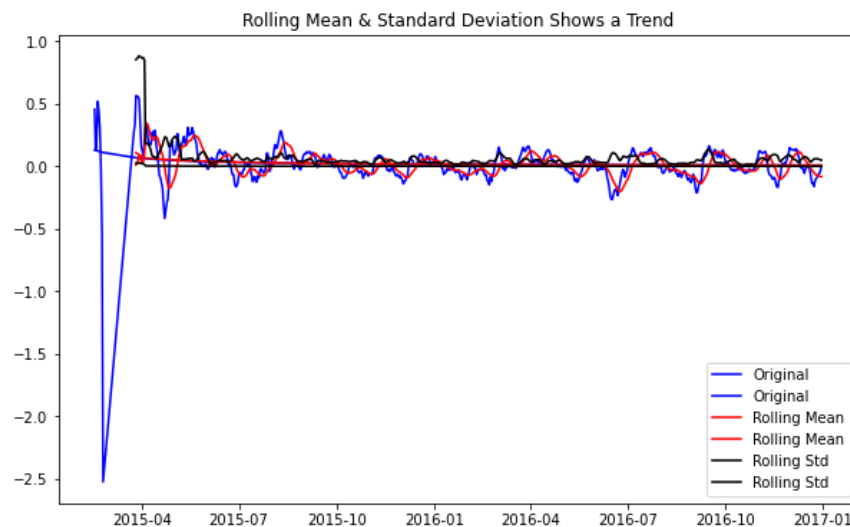
First, the dataset has a `.dropna` function run to make sure any not available values will not impede calculations. Then, the dataset is datetime indexed using pandas `date_range` function to allow calculations to properly show change over time. An arbitrary start date of 2015-1-01 is selected. A train set dataframe is created containing the first 701 daily measurements, and a test set dataframe defined as the 30 most recent measurements.



Analysis Findings

Seasonal Component

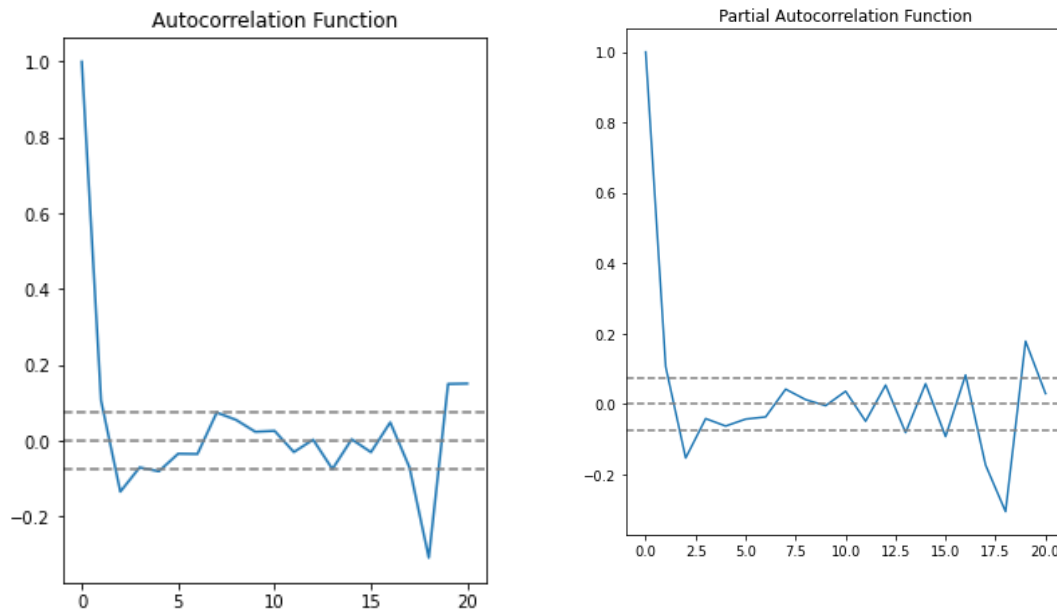
The cyclical nature of the seasonal plot confirms that the revenue dataset is seasonal.



Trends

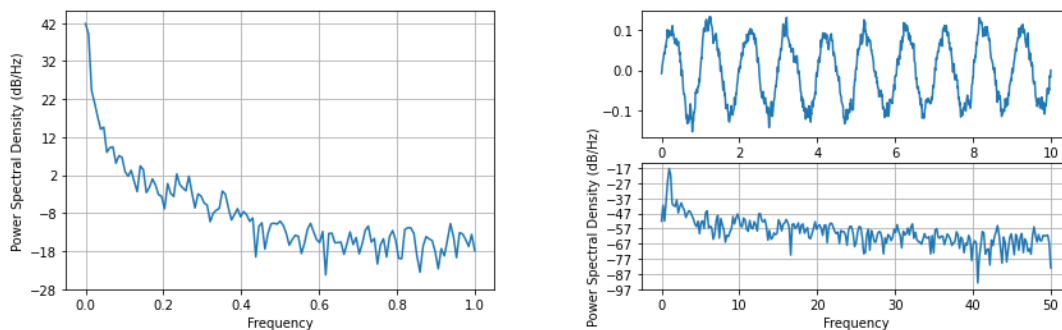
The trends plot indicates that there are trends relevant in the dataset. The chart visualizing rolling mean and standard deviation versus the original dataset also seems to visually indicate clear cyclical highs and lows

within the data. During the first few months, likely the hospital's ramp up period, the data is much more difficult to find these trends in, but once things stabilize after a few months these trends appear very consistent.



Autocorrelation Function

Both the autocorrelation function and partial autocorrelation function plots show that after the ramp up period, revenue values grow to be relatively consistent and predictable. Autocorrelation is a measure of how well a current value correlates with the prior values in the time series. Partial autocorrelation is calculated by finding the correlation within the transformed time series as before. Stable correlation within the range of -1 to 1 is seen for most of the observed time period.



Spectral Density

The spectral density graphs show a clear seasonal trend as there are many recurring cycles.

```

...:
...: from pmdarima import auto_arima
...: import warnings
...: warnings.filterwarnings("ignore")
...: stepwise_fit=auto_arima(df['Revenue'], trace=True, suppress_warnings=True)
...: stepwise_fit.summary()
Performing stepwise search to minimize aic
ARIMA(2,1,2)(0,0,0)[0] intercept : AIC=883.277, Time=0.75 sec
ARIMA(0,1,0)(0,0,0)[0] intercept : AIC=1015.972, Time=0.09 sec
ARIMA(1,1,0)(0,0,0)[0] intercept : AIC=881.359, Time=0.11 sec
ARIMA(0,1,1)(0,0,0)[0] intercept : AIC=906.199, Time=0.11 sec
ARIMA(0,1,0)(0,0,0)[0] intercept : AIC=1015.481, Time=0.05 sec
ARIMA(2,1,0)(0,0,0)[0] intercept : AIC=883.300, Time=0.20 sec
ARIMA(1,1,1)(0,0,0)[0] intercept : AIC=883.314, Time=0.19 sec
ARIMA(2,1,1)(0,0,0)[0] intercept : AIC=883.348, Time=0.39 sec
ARIMA(1,1,0)(0,0,0)[0] intercept : AIC=879.982, Time=0.06 sec
ARIMA(2,1,0)(0,0,0)[0] intercept : AIC=881.911, Time=0.07 sec
ARIMA(1,1,1)(0,0,0)[0] intercept : AIC=881.927, Time=0.08 sec
ARIMA(0,1,1)(0,0,0)[0] intercept : AIC=905.166, Time=0.05 sec
ARIMA(2,1,1)(0,0,0)[0] intercept : AIC=881.947, Time=0.19 sec

Best model: ARIMA(1,1,0)(0,0,0)[0]
Total fit time: 2.376 seconds

```

Decomposed Time Series

This decomposed time series graph showing revenue over time illustrates that there are seasonal cycles present.

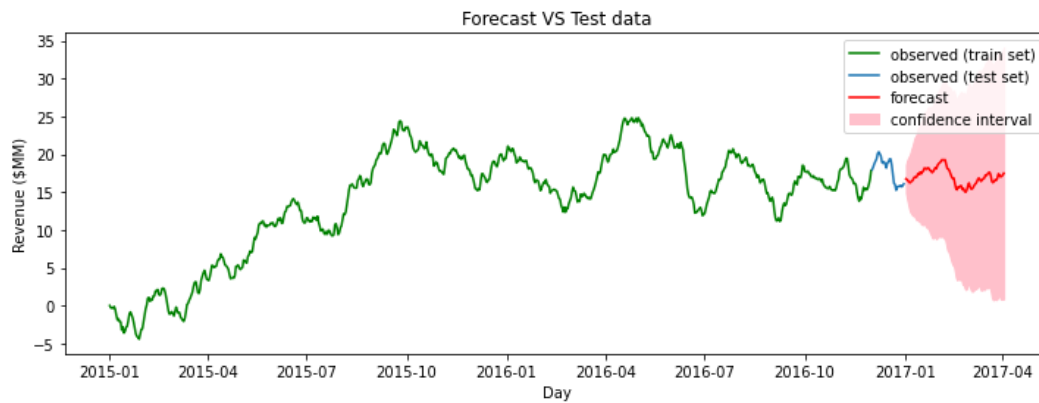
```

Out[6]:
<class 'statsmodels.iolib.summary.Summary'>
"""
                                SARIMAX Results
=====
Dep. Variable:                  y      No. Observations:              731
Model:              SARIMAX(1, 1, 0)  Log Likelihood              -437.991
Date:                Wed, 15 Oct 2025  AIC                        879.982
Time:                21:11:32          BIC                        889.168
Sample:              01-01-2015        HQIC                       883.526
              - 12-31-2016
Covariance Type:              opg
=====
              coef    std err          z      P>|z|      [0.025    0.975]
-----
ar.L1          0.4142     0.034    12.258     0.000     0.348     0.480
sigma2         0.1943     0.011    17.842     0.000     0.173     0.216
=====
Ljung-Box (L1) (Q):                0.02  Jarque-Bera (JB):                1.92
Prob(Q):                           0.90  Prob(JB):                      0.38
Heteroskedasticity (H):              1.00  Skew:                          -0.02
Prob(H) (two-sided):                0.97  Kurtosis:                       2.75
=====

```

Residuals

There is no clear pattern among the residuals of the ARIMA analysis, adding validity to the results.



ARIMA Model & Forecast

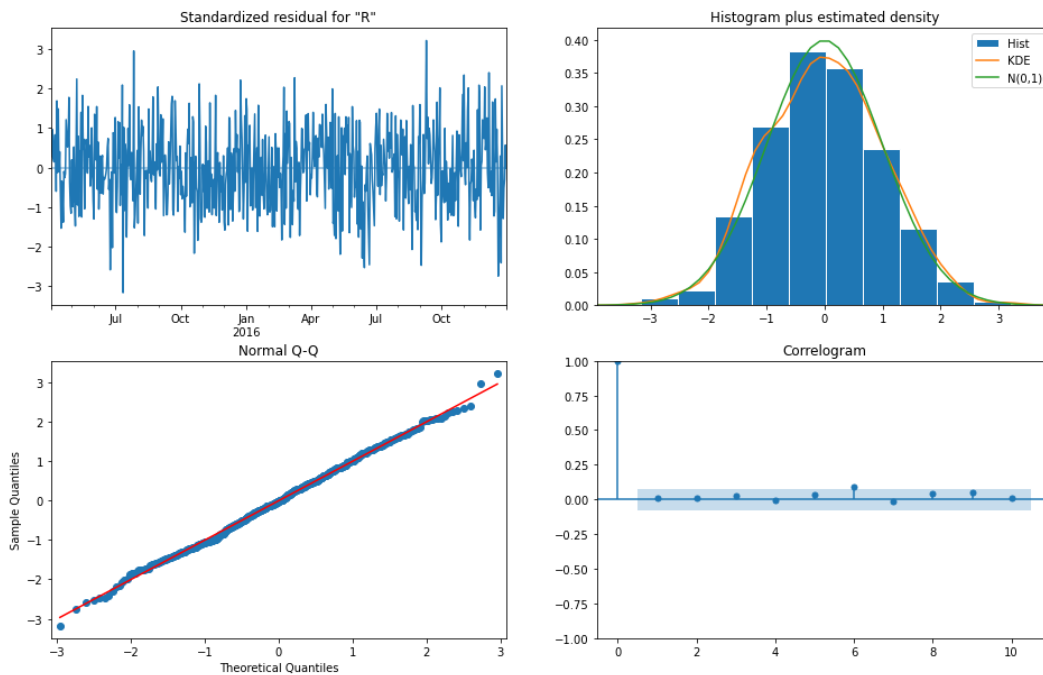
The implemented ARIMA model was built by selecting the recommended P, D and Q values of 1,1,0 from running the auto ARIMA model over original dataset. Ninety days has been selected for the forecast period.

Ninety days should give enough time to allow the company time to interview and go through the hiring process if new employees are necessary, ample time to prepare any supply orders, and give a strong estimate of the company's revenue to inform management if they will be able to expand facilities and equipment and begin those processes. Going beyond one quarter would begin to diminish the accuracy of the predictions while looking beyond the scope of current business operations.

Model Evaluation

```
In [91]:  
...:  
...: mae = np.mean(np.abs(results.resid))  
...: print("MAE", mae)  
MAE 0.452307939667293
```

First, the final ARIMA model was evaluated by calculating a mean absolute error score. This model scores 0.4523, indicating the average distance between the observed and predicted values is very close and indicates the model is making predictions accurately.



The final ARIMA model is also evaluated using a `results.plot_diagnostics` function, populating the above graphs. The standardized residual for R graph stays within a tight consistent range. The sample vs theoretical quantities line graph has the samples and theoretical values lined up very closely. These are more indicators that the model is predicting values closely to the observed values.

```
In [13]:
...:
...: from sklearn.metrics import mean_squared_error
...: from math import sqrt
...: rmse = sqrt(mean_squared_error(pred,test['Revenue']))
...: print(rmse)
1.7676801786810543
```

The revenue predictions generated using the test dataset can be compared to the actual revenue values in the test set to provide an RMSE of 1.76. A low RMSE score indicates that the mean difference between prediction values and actual values is low and indicates a strong model. This is another positive indicator for the implemented ARIMA model.

Forecast Visualization

The following graph first shows the two years of observed revenue, broken into test as the first 701 days in green, and train as the final 30 observations in blue. The graph then shows predicted revenue in red based on the forecast predictions provided by the ARIMA model built using the recommended P, D, and Q values. A confidence interval is also provided by running upper and lower confidence interval functions. An arbitrary start date of 2015-1-01 was selected for the observed data, so the observations end on the final day of 2016 to represent two years. The first three months of 2017, directly following the observations, are used for the forecast time period. Visually, the forecast data appears to remain stable and consistent with the ranges of the previous year and a half. The earliest six to seven months of the observed revenue dataset appear to be

a ramp up period so this is less relevant than the more recent data for forecasting.

Course of Action

Based on the ARIMA model forecast, it would be recommended that hospital management use the provided revenue forecast for the next quarter to plan hiring, supplying, and growth for project planning during this time period.